

CS & IT ENGINEERING



Computer Network-1

IPv4 Address

Lecture No. - 01

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Recap of Previous Lecture



Topic

DF Flag

Topic

Time-to-Live

Topic

Protocol Type

Topic

Header Checksum

*IPv4 Options
*ICMP



Topics to be Covered



Topic

IPv4 Address



ABOUT ME

Hello, I'm **Abhishek**

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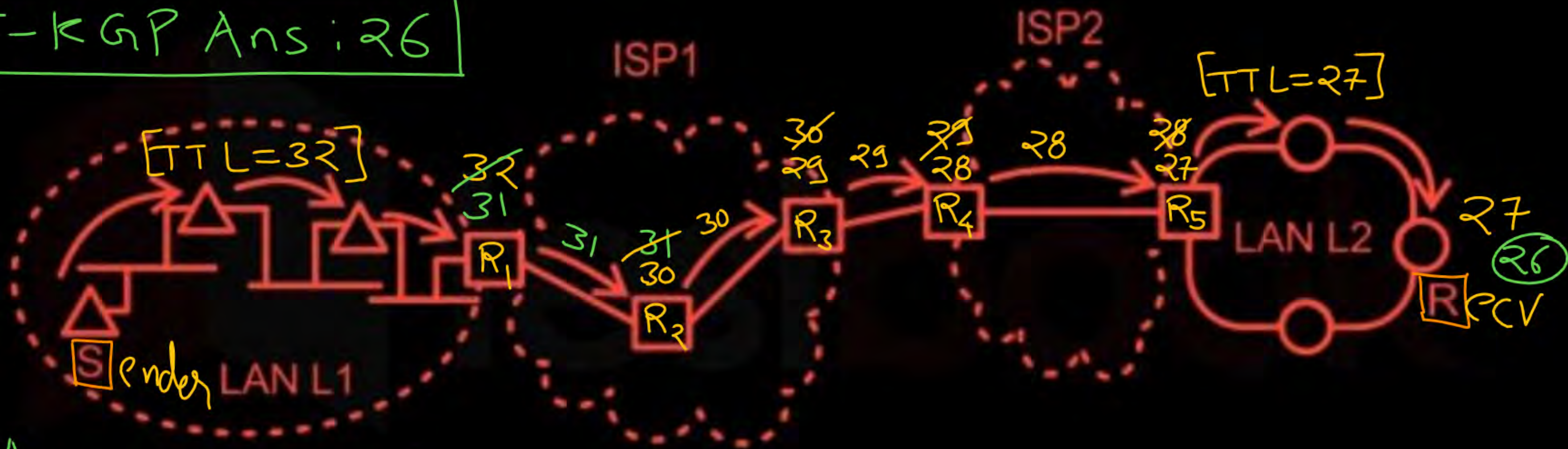
[NAT]

[GATE-2014][1 Mark]



#Q. In the diagram shown below L1 is an Ethernet LAN and L2 is a Token-Ring LAN. An IP packet originates from sender S and traverses to R, as shown. The link within each ISP, and across two ISPs, are all point to point optical links. The initial value of TTL is 32. The maximum possible value of TTL field when R receives the datagram is ____.

IIT-KGP Ans: 26



Ans = 26 or 27

[MSQ]

[GATE-2024][1 Mark]



#Q. Which of the following fields of an IP header is/are always modified by any router before it forwards the IP packet?

☒ A Source IP Address

☒ B Protocol Type

☒ C Time to Live (TTL)

☒ D Header Checksum

→ Always modify

Ans: C & D

[MCQ]

[GATE-2015][1 Mark]



#Q. Which one of the following fields of an IP header is NOT modified by a typical IP router ?



☒ A Checksum

☐ B Source address

☒ C Time to Live (TTL)

☒ D Length

Ans: B

[MCQ]

[GATE-2014][1 Mark]



#Q. Host A (on TCP/IPv4 network A) sends an IP datagram D to host B (also on TCP/IPv4 network B). Assume that no error occurred during the transmission of D. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D?

☒ (i) TTL

☒ (ii) Checksum

(iii) Fragment offset

☒ A (i) only

☒ B (i) and (ii) only

☒ C (ii) and (iii) only

☒ D (i), (ii) and (iii)



IIT-KGP Ans: D

[MCQ]

[GATE-IT-2004] [2 Mark]



#Q. In the TCP/IP protocol suite, which one of the following is NOT part of the IP header?

- ☐ A Fragment Offset
- ☐ B Source IP address
- ☐ C Destination IP address
- ☒ D Destination port number

Ans: D



Topic : IP Address



→ Unique identifier

→ Assigned to a device that uses Internet Protocol for communication

→ Logical Address (IP Address)

→ MAC Address (Physical Address) $\Rightarrow$ Hardware Add.
[48-bit size, NIC Card Address]



Topic : IP Address



→ IP address having two sections :

1. Network Identifier (Net ID) : x bits
2. Host Identifier (Host ID) : y bits

→ Size of IP address field = (x + y) bits





Topic : Network Identifier

- Unique network identifier (x bits)
[Range : 0 to $(2^x - 1)$]
- Assigned to each connected network
- Managed by IANA
[Internet Assigned Numbers Authority]
- Total number of networks (worldwide) can be : 2^x

[Ignore Reservations]



Topic : Host Identifier



- Unique host identifier (y bits) per network
[Range : 0 to $(2^y - 1)$]
- Assigned to each connected host in the network



Topic : IP Protocol



Two IP versions :

1. IPv4 ✓

→ IPv4 Address (32 bits)

→ 4 bytes [$x + y = 32$]

[2^{32} IP Addresses]

2. IPv6

→ IPv6 Address (128 bits)

→ 16 bytes [$x + y = 128$]

NAT

[2^{128} IP Add.]



Topic : IPv4 Address Representation

→ Size of IPv4 Address field = 32 bits

→ Two representation of IPv4 address :

1. Binary Representation [32 bits]

2. Dotted Decimal Notation



Topic : Dotted Decimal Notation

→ 4 Octet, separated by period (.)

→ P.Q.R.S $\longleftrightarrow$ 32 bit

→ Range value of every octet is 0 to 255

[8-bit Range : 0 to $(2^8 - 1)$]

0
.
.
.
.
.
.
.
255



Topic : IPv4 Address



→ IPv4 address are categorized into two type :

1. Classful IPv4 Address

2. Classless IPv4 Address ✓



→ Number of bits for network id field are assigned 'Dynamically'

→ Classless IPv4 Address : $P.Q.R.S / x$

↑ Prefix

→ Prefix / Slash notation [x] represent “number of network id bits”

start x bits $\rightarrow$ fixed

$(32-x)$ bits $\rightarrow$ host ID



Topic : Network Address

- Special IP address
- It can not be an IP address of any host in the network
- Used to represent a network

NetID field = As Assigned

HostID field = All Zero Bits



x bit

y bit

Example 1 :-

[MCQ]



#Q. Consider a class-less IPv4 network where network id bits assigned to this network are "1100 1100 1010", identify correct decimal representation of network address (special IPv4 address) which is used to represent this network?

☒ A 204.144.0.0

☒ B 204.160.0.0

☒ C 212.144.0.0

☒ D 212.160.0.0

Ans: B

Network Address =

11001100 10100000 00000000 00000000

12 bit NetID 20 bit HostID

204.160.0.0



Topic : Modes of Transmission

→ Three modes of transmission in IPv4 :

1. Unicast

2. Broadcast

3. Multicast



Topic : Unicast Transmission

- One-to-one transmission
- One host is sender and one host is receiver

Source IP Address = [Host IP Address]

Destination IP Address = [Host IP Address]



Topic : Broadcast Transmission

→ One-to-All transmission

→ One host is sender and all host (of a network) are receiver

Source IP Address = [Host IPv4 Address]

Destination IP Address = Broadcast IPv4 Address

→ Special IP Add



Topic : Broadcast Address

- Special IPv4 address (32 bits)
- It can not be an IP address of any host in the network
- Two types of IPv4 Broadcast Address :-
 1. Limited Broadcast Address
 2. Directed Broadcast Address



Topic : Limited Broadcast

→ Special IPv4 address (32 bits)

→ Used to broadcast a packet (to all hosts) of a network-segment
[in which transmitting host belongs]

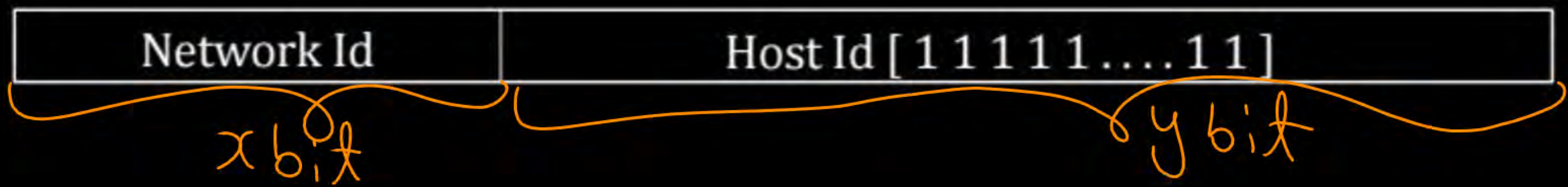
→ Limited Broadcast Address : 255 . 255 . 255 . 255
[All 32 bits are one]



Topic : Directed Broadcast

- Special IPv4 address (32 bits)
- Used to broadcast a packet (to all hosts) in a targeted network
- Network Directed Broadcast Address

<u>NetID field</u>	=	[As Assigned]
<u>HostID field</u>	=	<u>All One Bits</u>



Example 2 :-

[by default: classless]

[MCQ]



#Q. Consider a class-less IPv4 network where network id bits assigned to this network are "1100 1100 1010", identify correct decimal representation of broadcast address (special IPv4 address) which is used to broadcast a packet in this network?

- ☒ A 204. 160 . 0 . 0
- ☒ B 204. 160 . 255 . 255
- ☒ C 204. 175 . 0 . 0
- ☒ D 204. 175 . 255 . 255

Ans: D

Network Directed Broadcast Address =

11001100 10101111 11111111 11111111

12 bit NetID 20 bit HostID

204.175.255.255

Example 3 :-

[MCQ]



#Q. Consider a class-less IPv4 network, the network address (special IPv4 address) of this network is "145 . 75 . 80 . 0" and size of network identifier field is 22 bits, which one of the following can be a broadcast address (special IPv4 address) of this network ?

- A 145 . 75 . 80 . 0
- B 145 . 75 . 80 . 255
- ✓ C 145 . 75 . 83 . 255
- D 145 . 75 . 95 . 255

Ans: C

Network Address = 145.75.80.0

10010001 01001011 01010000 00000000

22 bit NetID

10 bit HostID

Network Directed Broadcast Address =

10010001 01001011 01010011 11111111

22 bit NetID

10 bit HostID

145.75.83.255

Example 4 :- [H.W.]

[MCQ]



#Q. Consider a class-less IPv4 network, the broadcast address (special IPv4 address) of this network is "175 . 175 . 175 . 255" and size of network identifier field is 22 bits, which one of the following can be a network address (special IPv4 address) of this network ?

- A 175 . 175 . 0 . 0
- B 175 . 175 . 160 . 0
- C 175 . 175 . 172 . 0
- D 175 . 175 . 175 . 0

Example 5 :- H.W.

[MSQ]



#Q. Consider a class-less IPv4 network, the network address (special IPv4 address) of this network is "145 . 75 . 80 . 0", which is/are can be a broadcast address (special IPv4 address) of this network ?

- A 145 . 75 . 80 . 0
- B 145 . 75 . 80 . 255
- C 145 . 75 . 95 . 0
- D 145 . 75 . 95 . 255



2 mins Summary



Topic

Network Address

Topic

Broadcast Address

Topic

~~Network Size~~



THANK - YOU

